

# The Regulation of Complex Organized Systems

Michel Crozier and  
Jean-Claude Thoenig

In this study French local government is analyzed. The analysis shows that French public affairs at the local level are managed by a complex, stable system of groups and institutions. The analysis also lends support to the contention that interorganizational relations are regulated by a complex, and more or less organized, system rather than by an interorganizational network. •

The relationship between organizations and their external environment has long been considered a sociological problem to be explored either through a culturalist approach derived from social anthropology or through an approach based on political economics or history. The heuristic value of these approaches is limited because they are based more or less implicitly on the assumption that there is a homologous relationship between the specific organization studied and the global society. They also suggest a deterministic view of social change. Not satisfied with these older approaches, organizational sociologists have adopted alternative approaches. One has been to apply the modes of reasoning and the methods of analysis employed in the study of the relation between the units or members of an organization vis-à-vis the relation between the organization and its environment. This new approach became possible in the early 1960s, when real progress was made in understanding the organization as an open system (Emery and Trist, 1965; Katz and Kahn, 1966).

Implicit in this new approach is the recognition that the relationship between an organization and its environment is based on dependence. Levine and White (1961) showed that the organization maintains relations with the outside world because of its dependence on the scarcity of available resources. Litwak and Hylton (1962) stated that, in fact, the dependence is reciprocal; the organization acts on the environment at least as much as it is affected by it. Weick (1969) pointed out that within certain limits the organization can structure and mold its own environment. Thompson and MacEwen (1958) proposed four possible organizational strategies for acting on the environment. Thompson (1967) stressed the fact that the environment constitutes an uncertainty which the organization tries to control and reduce.

The environment exhibits different characteristics according to the situation. Hall (1972) made a distinction between the general and the specific environment of the organization. Duncan (1972) showed that, in addition to an external environment, the organization can also have an internal one. Emery and Trist (1965) classified the environment according to four different types based on their complexity and fluidity. Burns and Stalker (1961), Aiken and Hage (1968), and Lawrence and Lorsch (1967) emphasized that the internal structures of the organization may vary radically depending on the nature of the environment.

A turning point was reached when the concept of environment was conceived as a universe of specific, differentiated units which must be defined. Evan (1966) proposed the concept of the organizational set which should identify and specify the organizations which form the particular environ-

•  
An earlier version of this paper appeared as "La Regulation des Systemes Organisés Complexes." *Revue Francaise de Sociologie*, janvier-mars 1975-XVI-1: 3-32.

ment of the focal organization. Warren (1967) emphasized that the environment is a structured field consisting of a series of organizations. Turk (1970) specified that this field constitutes an interorganizational network with particular characteristics and that it exists by virtue of some degree of social integration. Crozier (1972) showed that this network can be considered as a more or less structured organized system and that it can be analyzed as such by the methods developed by the sociology of organizations. Aldrich (1974) adopted the same idea. In this perspective, the concept of environment became a residual category identifying all that does not belong to the network.

As Benson (1974) noted, however, contributions made in the literature are more conceptual than empirical. They take the form of abstract typologies, stating theoretical conditions or critical programs rather than proposing empirical analyses or testing the concrete data of the new approaches. And even when such empirical analyses are set forth, they are comparatively weak and quite inadequate in view of the scientific objectives stated. The description outweighs the analysis. Causality is inferred, rather than demonstrated. The most commonly used methods are, on the one hand, the study of communication flows and resource exchanges, and, on the other, the comparison of formal characteristics of the organization—structure and output—and the types of environment. The problem of change remains untouched since the social universes studied are treated in a static manner.

The present article is designed to make an empirical, as well as a methodological, contribution. Its purpose is to bridge the gap between theory and empiricism in this new field of complex and partially organized interorganizational systems. Its objective is to show, through a concrete case study, the advantage of one mode of reasoning in understanding the subtle regulations which govern interorganizational milieus. The concept of a complex and more or less organized system seems preferable to the idea of an interorganizational network. For a system means regulation, integration, and homeostasis. To claim that an organization is part of a larger system is to require sociological proof that the system means more than the sum of its parts, and that there is a collective game which exists independently of the individual games played by each of the organizations.

To test such an approach, a special case of collective decision making in France is analyzed. This case shows that public affairs at the local level are managed by a complex, but stable and organized, system of institutions and groups. Such an analysis relies largely on the results of research on the system of power at the level of the *département* (departments) and the role of political, economic, and administrative élites (Thoenig, 1975).

The research was directed by Jean-Claude Thoenig. Between December 1973 and March 1974, a survey was carried out in three departments—Somme, Allier, and Hérault. Five hundred twenty-seven officials were interviewed who belonged to the main institutions—state bureaucracies, economic association, and so forth—or who were mayors or other elected officials in these departments.

In addition, this case also draws upon the results of other researches previously made by the fellows of the *Centre de Sociologie des Organisations* in Paris under the direction of Michel Crozier, particularly by Pierre Grémion (1976).

## **COLLECTIVE DECISION MAKING AND LOCAL GOVERNMENT IN FRANCE**

Various approaches can be used in the study of collective decision making in public affairs. Some of the paradigms frequently applied to the study of local government, particularly in and about France, are administrative science, political sociology, and class conflict theory. A different approach, that of system analysis, is suggested here.

### **A System Based on Interdependence**

At first sight, a coordinated style of management of public affairs at the local level seems impossible in France. Numerous institutions and groups are in charge of local government at each territorial level. There are two basic levels of local government: the *commune*—there are approximately 34,000 communes in France—and the department—there are 96. In addition, other public authorities exist which are intermediaries between the commune and the department, but which do not have much importance, such as the *arrondissement* and the *district urbain*.

The commune is managed by a municipal council elected by the population. The municipal council is headed by a mayor who is chosen from its members. Each commune, whether a city of a million inhabitants or a small village, is organized in the same way.

The department is governed by a legislative body, the *conseil général*, elected popularly, whose major task is to vote the budget for public works, welfare, education, and the like.

A complex mosaic of private and public authorities act as representatives of economic and social interests, such as the chambers of commerce, the chambers of agriculture, workers' and farmers' unions, employers' associations, the political parties, and citizens' committees. Even the central state intervenes directly in local affairs and it does so through several autonomous bureaucracies. Each state ministry has its own field agencies located in each department. Formally, one state field agency, the *préfecture*, has played a particularly important role and more specifically, its director, the *préfet* (prefect). There, high public servants are appointed by the state to be the representatives of the central government in the departments. Simultaneously, they are also the executive branch of the general council. Field agencies of the state bureaucracies also play a double role; they implement the decisions made by their ministry in Paris and they check the decisions made by local authorities to determine if these decisions conform to the technical rules of the state. Coordinative measures are taken regularly by the central government and its technocrats to improve the level of cooperation, but they are hardly adequate, as shown in the case of the 1964 reform, which was supposed to increase the power of the prefect as the coordinator of public management at the department level (Grémion, 1966). Everybody complains of

isolation from other units; fragmentation is a basic characteristic of local government; and the information flow is very low. Historic rivalries, conflicts of interests, and reluctance to cooperation are common.

Despite the lack of coordination and communication, however, the activities of the numerous institutions, organizations, and groups are interdependent. They are all involved in the same game, although, to a large extent, it is a game of defense, protection, and noncommunication. If there is steady resistance to coordination and formal adjustments, it is not because the players do not know one another, nor because they are far from one another, but because they belong to an informal, but complex and stable, system which has other forms of regulation. This system may be only a partly explicit one, but it is nevertheless efficient.

The first sign indicating the existence of a game common to these organizations is the presence of a series of paired relationships between some of the key participants, as Worms (1966) noted in his skillful analysis of the relations between prefects and mayors of middle-size cities. The apparent hostility of mayors toward their prefects and the criticism that prefects direct towards their mayors conceal a deep complicity; efficient prefects know how to listen to their mayors and efficient mayors are naturally attentive to their prefects. Their activities complement each other and are comprehensible only if one studies the pair formed by the two partners. Although the mayors must take the initiative, they allow themselves to be prompted by the prefects, provided that the prefects agree to let the mayors receive the recognition and the prefects take their mayors' needs and wishes into account. The subtlety of the relation can lead to misunderstandings and problems in communication and it does not facilitate rational and formal coordination. There is interdependence, however, and to understand the results of this joint activity, it is more important to understand the functioning of the system than to focus on the a priori preferences of the decision makers.

The case of the prefect and of the mayor is not an exceptional one. There are many other cases in which state bureaucrats and local élites have established close interpersonal relationships. Every director of the field agencies of state ministries has some local partners, such as political and economical officials; every local official has privileged partners within the state bureaucracy. The director of the field office of the ministry of agriculture, located at the department level, for instance, and the director of the local chamber of agriculture are one such pair. Another pair is composed of the mayors of small towns and the heads of local offices of the ministry of public works located at the infradepartmental level.

The present research suggests that, generally, communication is easier between state officials and local political and economic élites—even if formally they are independent and their respective institutions are rival—than is the case among themselves in their own organizations, where conflicts between hierarchical levels result in a low level of communication. To some extent these pairs of individuals belonging to different institutions are more integrated with each other than

they are to each of their respective institutions. Differences between state officials and local leaders and between bureaucrats and *notables*<sup>1</sup> are considerably less than they appear to be. In many ways, the bureaucrat is a *notable*. As for the local *notable*, he is often a better expert in bureaucratic affairs than his partner, the state bureaucrat (Grémion, 1975).

There emerges complicity among these groups based on the sharing of a mutual experience and complementary interests and on the adherence to the same collective norms. Such complicity is strong and enduring. The frequency of interpersonal relationships correlates with attitudes toward that institution to which each partner belongs. The more *notables* have a personal relationship with state representative, the less they are dissatisfied or critical about the state bureaucracies. Negative attitudes and aggressive criticism are common among local officials who do not have frequent and direct personal access to department-level state officials. To a certain extent, the network of state agencies that spreads over the national territory is a basic mechanism through which interdependence between apparently rival and even conflicting levels of government is achieved. State bureaucrats accept the fact that political, partisan, or parochial strategies are important dimensions in the games played by the *notable*. Local *notables* recognize the legitimacy of the norms and rationalizations the state bureaucrats use to develop their policies or justify their direct intervention in local affairs.

### General Rules of the System

As in any complex system, no decision can be made unilaterally by any one of the actors. Interdependence is particularly visible every time an institution has to act or make a decision. Parties must reach acceptable compromises and take into account the existence and the strategies of other actors.

In the French system of local government and bureaucratic action, a compromise cannot be negotiated directly by the parties immediately involved. It is brought about through the intervention of a third party, an external actor, an individual who does not belong to any of the groups to which the parties belong. Such a mechanism fulfills the functions of coordination and of integration, through which compromises, and thus decisions, can be made. Integration and coordination are maintained by an institution, or more precisely by individuals whose activity and whose sources of legitimacy are derived from groups other than those they seek to reconcile. Often, the coordinator imposes a preconceived solution on the parties, one which takes into account the different interests but is imposed without open negotiation. Such a solution succeeds even more when it is presented in the name of another rationality: technical imperatives, legal constraints, the general interest.

Each unit in such a system acts by itself through channels open to it, without worrying about its neighbors, without communicating with them—eventually by concealing itself from them—and without seeking to create a common front with them against the intervention of outsiders. Avoidance of cooperation among the units involved is the norm. Each

<sup>1</sup> The French term of *notable* has no convenient translation in English. The French *notable* is the local or regional person of influence who has been granted informal recognition by the state authority. The *notables* are members of the local élite whose prominence—because they have acquired a strategic position either by popular election or by representation of powerful interests—makes them a necessary partner for decision making. Mayors or general councilmen are *notables*.

seeks to resolve its own problems with the intervention of outsiders, especially those from a different group rather than by direct bargaining with its peer groups. Intervention of an external force is therefore crucial to coordinate the units and to allow them access to the center.

### **The General Rule in Practice**

To understand how this rule works, one has to study how each unit solves its decision-making problems. The French system is structured around the intervention of specific outsiders in the decision-making process of each unit. One can draw a somewhat distorted, but more lively, description by beginning with the central figure of the mayor of a village or of an average-sized town. Mayors are the only people who fulfill a function of integration and coordination in the community which they head. As a result, they are in an exceptional position of power. This power does not flow directly from their official position. Rather, it derives from the fact that the different social categories and interest groups in their communities, even if represented in the municipal council, are not capable of reaching compromises by themselves. It is the mayor who, perceived as uninvolved in any of the groups and therefore of another nature, makes decisions for the general good (Kesselmann, 1966). In practice, mayors select their municipal councilmen more than the municipal councilmen elect them. Mayors take care to include in the list submitted to the constituency representatives of various interest groups who are, in fact, their hostages. Municipal elections are highly personalized in mayoral elections, even if the mayor is not elected by the population but by the municipal council. No external group or leader can really organize a coalition of interests against the mayor once in office. Interest groups intervene but, even if they are capable of putting pressure directly on the mayors, they have great difficulty in manipulating their counselors against them. Should they succeed, disorder begins and the community is torn apart.

But these powerful mayors are nevertheless handicapped, perhaps more than others, when they have to take positive action. They have limited monetary resources at their disposal, and no recourse to an independent technical expertise to carry out their undertakings. They cannot obtain sufficient resources unless they receive a subsidy from the state. To get the subsidy they must present good technical and financial dossiers, ones developed by such field officers of state bureaucracies as the ministry of public works or of finances. The administration's bureaucrats have a *de facto* monopoly of expertise in matters such as public works, housing, public finance, and traffic policy. They are the best advisors to ensure the success of any demand in these fields. They know how to plan a road project or to present a municipal budget so that they fit the state bureaucracies' standards. Most of the local officers of the state do not hesitate to give valuable advice to the mayors.

Mayors are largely dependent on state bureaucrats located in field agencies. Without them, they can accomplish nothing. The head of the local subdivision of the ministry of public works, for instance, determines the public works policy of the mayor, with the help of the state tax-collector who oversees

the amount of the community's debts and resources. Some state bureaucrats even act as if they were part of the executive branch of the commune. Mayors ask them to build city facilities or maintain the municipal network of roads. These bureaucrats act as the mayors' officers and they receive fees in return (Thoenig, 1973). In exchange for such a dependency, the mayor is guaranteed supposedly sound management. Community policy is not only the responsibility of the mayor, but it becomes a field in which state bureaucrats play an important role and intervene as if they were the actual bosses of the commune.

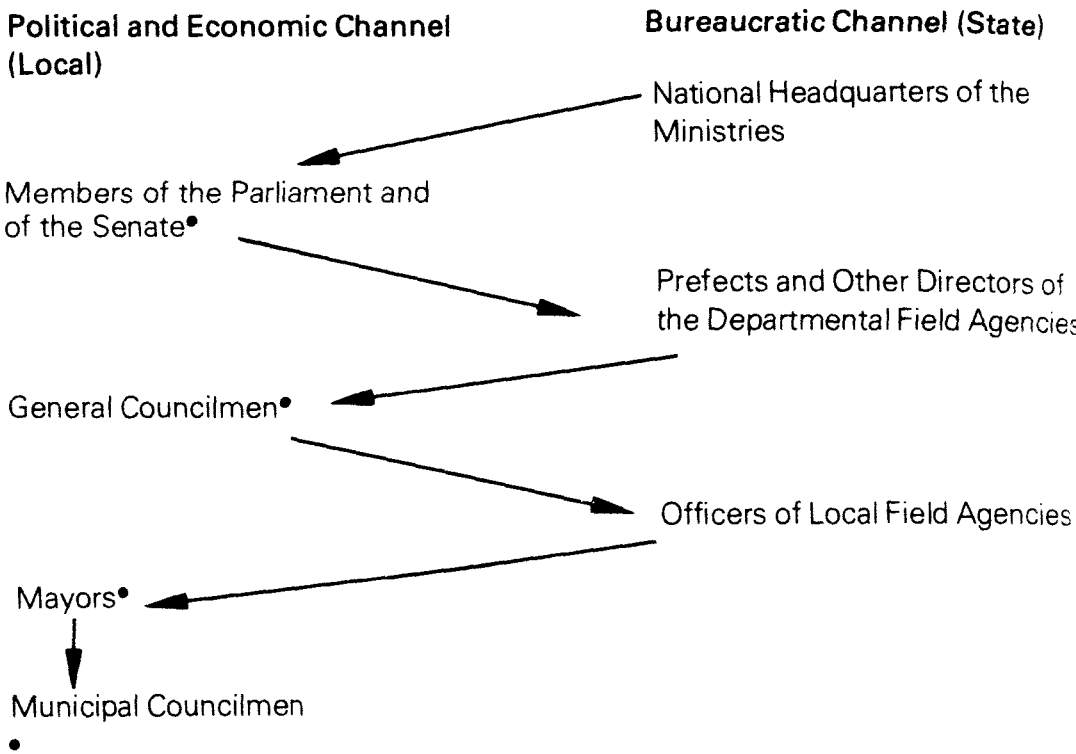
The fact is, however, that the mayors feel that the state bureaucrats who intervene in their municipal affairs are not a threatening danger, but a positive help. Their help is needed if the mayors want to succeed and be perceived by the population as efficient and prestigious. On their side, state bureaucrats tend to have their own local policy which they implement and impose all the more as they are welcomed by the mayors. They also receive such benefits as consulting fees and power in the community.

The system is by no means closed. Power is not shared by the state bureaucrats of the field agencies alone. Contrary to a mythical image of French centralization, they do not receive directives from their hierarchical superiors in the departmental agencies or in the headquarters in Paris. They first take into account the wishes, orientations, and strategies of the local political class. They dominate each mayor individually but, at the same time, they are prisoners of their clientele, the elected officials. Their autonomy suffers, therefore, some important limits. They cannot afford to antagonize unduly those whom they influence, because of the risk of destroying their right to intervene. They adopt a political approach. They are conditioned by the environment in which they govern. It is particularly important for them to take into account the strategies and goals of the most powerful political and economic leaders at the local level, such as the general councilmen. They have to accept the leadership of these people when they intervene. Being, in their turn, members of a milieu in which the parties because of the fragmentation are unable to achieve compromises, the state bureaucrats at the local level are coordinated, in fact, by some of the most influential *notables*. The intervention of these *notables* in the inner affairs of the field agencies of the state administration is so strong that some hierarchical superiors of the state bureaucrats can interfere in the activities of their subordinates only by acting on a political level, by influencing the *notables* who control the policies of the state field agencies.

Hence, the process of power and decision making is neither hierarchic nor democratic nor contractual. It is an intersecting, zig-zagging process particularly conducive to shunning responsibilities. Deeply anchored in local life, it extends to the departmental level and even to Paris. The position of the prefect and the principal departmental directors on the one hand and the general councilmen on the other is similar to that of the mayors and the lower levels of the state bureaucracies. The game is identical in the sense that decisions which form the policies of the general council are drawn up by the representatives of the state. At the same time, the



latter must make allowances; they are strongly influenced by the political milieu, particularly by the two or three most important *notables* in the department who know how to play at partitioning different administrative channels and how to assert their positions as coordinators, setting themselves up as indispensable intermediaries between the department and the ministers in Paris. Overall vertical communication is very weak in such a system. Local state agencies and their headquarters and mayors and general councilmen operate at a very poor level of cooperation and communication. For political and economic officials and state bureaucrats at each level in each channel integration is achieved by individuals and groups belonging to other channels.



• *Notables* sometimes—but not always—occupy these positions.

**Figure 1. Cross-regulation mechanism.**

### The Exception to the General Rule

One basic characteristic of the system which transcends political opposition and geographical differences is the prevalence of the plurality of elected offices. A number of mayors accumulate the functions of a general councilman and of a member of the national parliament. More precisely, most members of Parliament and the general councilmen are simultaneously mayors. Plurality of offices is the absolute requisite for a successful political career. These mayors naturally play the role of informal leaders. They have attained a position of influence by the simple fact of occupying offices which carry with them varying degrees of influence. Plurality of offices cannot be understood according to electoral and partisan reasoning. It exists for organizational reasons, with strategies played by local groups within the cross-regulation system.

If local political figures can hold three or four offices, whose incumbents communicate poorly among themselves, their positions become enormously strengthened. They can play several games simultaneously and are certain of winning since, inasmuch as these games are interdependent, the practical knowledge they develop in one role is the decisive



factor for success in the other. Elected officials who handle all the functions and base their negotiations with ministers, departmental directors, and local representatives upon this are in a much better position than their colleagues who can communicate with each public official only through indirect pressure on their indispensable intermediaries. Therefore elected officials are in a position to exert their leadership as political figures in a more acceptable manner. They are able to reward their friends without punishing their enemies too much and without rousing the animosity of the public. They can obtain satisfaction almost without asking for it and, with no effort on their part, serve as an extremely powerful integrative force. Intermediaries are important in a system in which favors are essential, but in which favoritism is greatly disapproved by the political culture.

However, elected candidates who hold several offices at the same time are not antagonists of the administration; on the contrary, influence depends largely on personal access to the most important officers in the bureaucratic channel. They become accomplices and know how to influence state representatives so that they take into account their desires and their influence. Holding several offices simultaneously may be an exception to the general rule of cross-regulation but in fact it fulfills an important function in the system. It reinforces the whole structure of relationships and the rules of the game because it offers a possibility of fluidity and homogeneity in a system that is otherwise rigid and fragmented.

### The Functions of the System

The network of organizations which intervene in local management of public affairs is a complex system structured by the interpenetration of different channels. These include the administrative channel of the central state, the world of the local communities and their political bodies, and economic associations. They are intertwined according to a general rule and are connected even when there is an exception to that rule. Each channel regulates the other one as well as being regulated by it. Between each channel there is communication and integration through the hierarchy of the alternate channel. Hierarchy and leadership are never direct. Substitution is essential in this system of cross-regulation.

The corollary to this process of cross-regulation is partitioning. Each unit is sustained and maintained by pressures from the system and each totally disregards other units. It is advantageous for a unit to communicate as little as possible with others in its own channel that are potential competitors or superior to it, since they could reduce it to a position of dependence. Its possibilities of success derive from a skillful use of the cross-control system through which it can bargain for influence. Thanks to such a system, direct confrontations are avoided. Formally, relationships are easy and friendly, but no one speaks to anyone and all cope by themselves. The result is isolation or atomization of the political fabric.

Such a system favors and even demands a strong concentration of power and privileges. The rarefaction of communication and the scarcity of passageways produce a kind of social rent for the few groups and individuals who control positions where communication and access to another level are possi-

ble. Influence and initiative are limited to a small group: the mayor, the *notable* who accumulates offices, the prefect and the president of the general council, and a few political figures with a national standing in Paris. At each level power and influence are polarized in the hands of two or three leaders who integrate all the units in the area (Grémion, 1975; Thoenig, 1975). The game is closed and secret. The system constantly produces exclusions. Nonparticipation is so well grounded in the psychology of the leaders that it seems consubstantial to their game. The members of the system fear public opinion and hide from the sanction of electoral suffrage as much as it is possible. Empirical evidence shows that the *notables* as well as the bureaucrats perceive ordinary citizens as lacking in self-responsibility, interested only in individual and parochial issues, and being reluctant to participate more in the management of public affairs. Citizen participation can only be self-destructive and the public is seen to follow demagogues.

Such a system is centralized. By centralization we mean a structural division of labor between the hierarchical levels where decisions are made and the levels where these decisions are implemented. Those who decide at the center are not those who are affected by the decision at the periphery (Crozier, 1964). But the decision makers are not free to do whatever they want and even if they wield great power, they are to some extent dependent on the hierarchical levels at the periphery. In fact they decide but they do not have the relevant information to make wise decisions. The system is centralized but there is poor circulation of information. Information is always mediated and transmitted from tier to tier by third parties. Hence, decision makers receive information already obsolescent or biased and those who execute the decisions have firsthand information but do not have access to the decision making.

The *notable* or the state bureaucrat is in fact a mediator, the defender of local interests, rather than a decision maker. If, for example, it is so important for a department to have a powerful leadership, to elect political figures, or to have a prefect who has direct access to the highest level in Paris, it is because these representatives serve as a pressure group in Paris. For the population of the village, the mayor is the one responsible for promoting the community's interests at higher levels in the system. The best mayor is not the one who achieves the most, nor the one who is most politically attractive; it is, rather, the individual who has best access to those who make decisions at the highest possible level. In fact, the honeycomb structure functions as the structure of access to the top. Cross-regulation creates relays through which demands and pressures can be introduced and processed (Grémion and Worms, 1970). The dependence of inferior levels on higher levels does not stem from the existence of formal hierarchical authority; it comes from the mere fact that the latter monopolize the promotion of the formers' interests in the right places. Such a mechanism is not a subterfuge. Indeed, the regulating force identifies itself with the interests in its control. Both the *notables* and state officials perceive the field agencies of the state ministries, as being representatives of local interests in the state apparatus, as well as

being the representatives of the central state in the local community.

Paradoxically, those who impose their will from the top are, at the same time, the representatives of the subordinate units they are supposed to control. The back and forth process of decision making, however, is not simple. The representation of the interests of one specific unit within one specific channel is managed by the leaders of another specific unit from another channel. This is essentially a process of intermediaries in which the intermediaries are indispensable and exact a high price for performing their functions. This means also that the main actors, the *notables*, whether elected officials or bureaucrats, are basically intermediaries who jealously guard the sources of their power, the monopolies of access and communications.

Mayors will not relinquish to anyone the responsibility for the city's relations with the *notables* of the bureaucratic, as well as of the elective or representative, channels. This makes them indispensable figures and enables them—if they are skillful enough—to outwit possible opponents who will always appear as lacking this basic expertise. Similarly, prefects jealously guard their personal contacts in Paris and the three or four most influential *notables* of the department are powerful only because they alone have access to the key political and administrative levels in Paris.

### Individuals and the System

At first sight, the French system of collective decision making is oppressive and inefficient. As a matter of fact it causes considerable frustration and strong criticisms among its members, as well as among the population. Yet nobody really works to overthrow it and it continues to be rather well accepted.

To understand such a situation, the behavior and attitudes of the members of the system have to be reconsidered as they are expressed in day-to-day relationships. In such a perspective, it is evident to the external observer that the system is in fact a less rigid and a more human way of governing public affairs than expected. Despite the unequal distribution of privileges and lack of participation, it maintains strong equality and a relevant degree of autonomy for each of its members. Moreover, it is pressure from its members that reinforces and stabilizes the system.

The system is oppressive and authoritarian but, at the same time, oppression is anonymous and authority impersonal. One or two particular individuals cannot arbitrarily impose their personal will and dictate their own rules to others. The game is collective and does not depend on the strategies of particular persons; it is a consequence of a system which imposes its general rules and norms on all individuals, even the most powerful. Its regulation depends on the stability of a set of relationships between units and not on the control over specific individuals. Therefore, oppression is impersonal and diffused. Consequently, it is tolerable. Simultaneously and independent of specific stakes they have in the collective game, individuals can form friendly relationships with others belonging to different units. The conflict is one between

roles, not individuals; hence, it does not hinder individuals from communicating in other matters.

The system is based on dependence and passivity. So while conceding their initiative and independence, individuals obtain other kinds of advantages. Inside each channel, the cross-regulation mechanism permits the superior echelon to evade the too direct pressures from the lower ranks, while the latter to choose not to submit to the too obvious domination of the former. Thanks to the intrusion of individuals from a completely different level and channel, the sources of conflict between peers in the same channel are suppressed by avoiding the face-to-face negotiation of compromises. All members are guaranteed they will not lose even if they err. No sanction is possible since the individual responsibilities are neither evident nor direct. The blame always falls on the outside regulator, or better still, on the system. Each individual is not held accountable but autonomous, and protected. All *notables* are guaranteed some measure of success, even if they have made no effort. Success is highly personalized and individual charisma is an important product of this collective game. Those who wish to escape the system, may do so, but at a considerable risk. If they take the initiative, they immediately become answerable and are no longer protected by the system in which they have become deviants because they do not play the collective game. Hence, for the individual the best strategy is to do as little as possible. The person who starts last has an advantage over the one who wishes to start first. The race is one of slowness. Apathy and nonparticipation constitute a rational strategy. The *notables* mobilize themselves only after the system has given the green light. But once it is given, they hasten to the point where they can no longer be stopped. Once the system is set into motion, no one is able to control it.

The system causes exclusion and favoritism. The rule is most useful as a reference for the possible exceptions. The exception, however, and the privilege it offers to those who take advantage of it, provides jealousy. Each one tries to have an exception, as if the system existed in order to legitimize exceptions to its own rules. The *notables* expect the state bureaucrats to adopt the role of rule breakers, of exception-facilitators. In the end, everyone manages to have his privilege but, at the same time, everyone is somewhat excluded. In short, the system produces exclusion and privileges which create discontent and uneasiness. But all is done in a sufficiently well-balanced manner, so that complaints against the system do not go beyond the acceptable threshold, that is, do not create imbalance in the personal benefits each member expects to receive.

In such a system, centralization itself is maintained and increased by the impulse of those excluded from decision making. The local community delegates all political power to one person, the mayor, because this is the best means for the community to be well represented at high levels, while being also protected from external pressures. By the same token, if the rural townspeople agree so frequently to entrust the office of counselor to mayors in the major towns in their districts, it is because of the power they exert through their

personal relations or their social origins. The prefect who represents the central state in the department is perceived by the *notables* not as an unbearable tyrant, but as the representative of local interests vis-à-vis the central state.

The local community feels better protected inasmuch as the central government is more directly involved in its growth and development. The intervention of Paris in local affairs is not just desired, but demanded. One complains of Paris not because it does not leave initiative to local people, but because it does not intervene enough. The intervention of the center in the affairs of the periphery is not a threat but an advantage. The advantage is won because of the efforts of the *notables* who have the easiest access to the highest level possible. Not to be represented by efficient mediators is a major handicap. In this case, the alternative is for local leaders to politicize the problem, to make an appeal to public opinion in the hopes of being heard at the top. This strategy is dangerous, however, and the chances of success very uncertain. It is much better to go through the cross-regulation mechanism.

Therefore, collective decision making in public affairs is structured by a very stable system. Everyone is a prisoner of it, but each member gains something in return. The advantages are financial and technical, as well as being symbolic, political, and even affective. The consequence of all members trying to win their own game and to beat the system is that everybody upholds the general rule of the collective game.

### **The Dynamics of the System**

The interorganizational network of local management of public affairs is a homeostatic system. Faced with new demands or the changes of society, it hardly adapts itself. On the contrary, it puts pressure on each of its parts in an effort to maintain itself. For instance, in the last two decades, as a result of rapid and important socio-economic transformations in France, new social groups have appeared and the general population has expressed more forceful demands for participation in public affairs. Rather than making itself more open, the system has closed in upon itself. Rather than stimulating more initiative from the periphery and broadening its base, it has sought to preserve its secretive and restrictive practices. The *notables* and the state bureaucrats have remained blind to its performances as a whole and to external pressures which could threaten the general game. They are so well socialized by the system that they are incapable of adapting it. No solution other than the traditional ones can move them. For instance, they oppose decentralization of government—seeing it as a dangerous policy. Participation causes politicization which is perceived by the political, as well as the bureaucratic, *notables* as the worst possible state of affairs. Indeed, the sole solution that they accept and implement consists in narrowing the number of members and of making the system more rigid. Even when strong pressure for change comes from the outside, as in the case of the reforms passed by the central government in March 1964, the *notables* and bureaucrats are able, as members of a system, to alter and radically change the goals and consequences of the central government (Grémion and Worms, 1970). The creation of the region, conceived of as a level of integration

superior to the department, did not bear fruit. On the contrary, it reinforced the department. Elected officials, using the plurality of offices, have succeeded in taking control of the region and have reinforced their privileges and their importance as regulators. Prefects of departments are stronger than they were before because, as intermediaries between each department and Paris, they can beat the region. The game today is even more restricted and exclusive than it was a few years ago (Crozier *et al.*, 1975).

### The Case of the Cities

Recent socio-economic changes—massive industrialization and urbanization—however, have been so important that the system as a whole could not help but adjust to these pressures for change. When its equilibrium is in jeopardy, its members react in a very specific way. They prefer to keep the structure of the system as it is, even if they have to pay a price. There can, for instance, be a general decrease of their influence in French society and a sharp restriction of the domain under their control. They prefer to have certain tasks taken away from the system rather than face threats to the present equilibrium. This has been the case with the massive growth of urban areas and large cities in the 1960s.

Traditionally treated as an exception to the rural rule, large cities have developed rapidly with the largest population and the greatest needs. Rather than making the necessary adjustments, members of the local system—*notables*, bureaucrats, and economic leaders—fell back upon the traditional game and allowed the large cities to elude the system even if, in turn, they had to resign themselves to being no more than the councils of small towns and villages. An autonomous new system for administering urban problems evolved under the sponsorship and control of the central government from which the old *notables* painstakingly hold themselves apart. As a consequence, there developed two different administrative networks of public management at the local level, one for rural France, the other for the metropolitan areas (Thoenig, 1973).

Generally, the system tends to disassociate itself from problems which it finds too threatening or insolvable, and concentrates on the problems with which it is capable of dealing. In this way its members can still feel that all is going well and perhaps getting better, even if the system controls less and less substance.

Unlike the case of the marginal player who refuses the protection of the system, the fact that urban affairs are excluded is not perceived as a disadvantage by the large cities. On the contrary, not only do their mayors evade the system, but, because they are excluded, they draw much more substantial benefits than they could have obtained within a system that is fearful of them. Their very rejection puts them in a favorable situation. First, they have direct access to Paris and Paris is obliged to play an active role in solving problems that have been caused by the cumulative lag in urban investments. Second, they have the immense advantage of unity of action. Like rural communities and small towns, large cities entrust their government to one man: the mayor.<sup>2</sup> But mayors of big

cities become much more powerful than their colleagues in small cities because they stand alone in dealing with a complex world of fragmented field offices and warring government bureaus. This loneliness makes it possible for mayors to serve as natural coordinators and necessary regulators of the system. This role is all the more crucial in urban affairs and metropolitan areas where the conflicting interests are more numerous and the bureaucratic apparatus of the state more complex. Big city mayors alone have direct access to Paris and have that contact *ex-officio*, without intermediaries. They are relatively free people in a world ensnared in networks of dependence and obligations. As a result of their role as regulators, mayors of large cities can exert influence and control over the territorial system from which they are excluded. They dictate their wishes to the local branches of state agencies. They act as regulators of neighboring communities, and the equal of the prefects—the prefects having lost their advantages as the natural spokesmen in Paris. The mayors can even be on bad terms with prefects without having to suffer negative consequences. Even in their dealings with Paris, the mayors are not completely dependent. In working with the multiple agencies which in Paris handle urban affairs, they are the sole representatives of their metropolis and the only ones in their cities who can take the initiative. They benefit from all the advantages of plurality without bearing any of the responsibilities.

### **Other Exceptions to the System**

Mayors of large cities are not the only people who are exceptions to the general system. Other groups such as farmers or industrialists are marginal or are excluded from the system and thus are in a position to beat it. The French system of local government conceals other systems, parallel arrangements which are set up outside of it and even working against it. But the systems are deeply connected and exist only by virtue of one another.

Groups and individuals struggle among themselves to seize the opportunity to develop such games and to occupy the positions which grant the privileges associated with these games. Despite its inflexibility, the basic system reveals certain dynamics and a genuine fluidity. The competition is keen and no one is assured of victory. The value of privileges is enhanced in a world where pressure toward egalitarianism is strong and constant. To hold several offices concurrently or to organize oneself in a corporate manner requires much time and effort. But it is also difficult to maintain the privileges which are the rewards for the plurality of offices or the corporate organization. Those who succeed are those who know how far they can go, those who create advantages for themselves without provoking too strong a counteraction. In such a system, it is essential to have an intuition of the accepted limits of the game to play these second-order strategies.

### **A SYSTEM APPROACH OF INTERORGANIZATIONAL NETWORKS**

Although stimulating progress is already suggested by studies made in fields as different as the theory of the firm (Lawrence and Lorsch, 1967; Emery and Trist, 1965; Dill, 1958); the



study of health, welfare, and education institutions (Aiken and Hage, 1968; Turk, 1970 and 1973; Clark, 1965; Litwak and Hylton, 1962); or the sociology of urban affairs (Warren, 1967); there is consensus among researchers that inter-organizational networks are specific levels of integration and interdependence between units. Turk (1970) underlined the fact that what must be studied are the collective rules and the processes of integration, the games of conflict and cooperation in such networks.

Such a strategy of research is stimulating not only in understanding interorganizational networks as specific phenomena, but in improving a general theory of organizations. Regulation is the basic mechanism of organizations. From that point of view, there is no difference in kind, but only a difference in degree between an organization and an interorganizational network. Both are social and human systems, more or less stable and structured, integrating various units, regulating their behaviors, and imposing a collective game on their members. While devising a paradigm of the regulation of organizational systems is a worthwhile way of developing a theory of organizations, research in this field, however, is slowed because there is a lack of alternatives to the two dominating trends. This is especially the case of the American tradition: organizational students tend either to try to build an abstract theory from which they deduce from a set of cybernetic axioms and derive normative prescriptions or to measure those empirical relationships about which one can get reliable data even if they are meaningless for understanding how the system really functions. Both these approaches now have lost any power to stimulate. Approaches can be developed which would not put the emphasis on the frequency of the relationships or on the formal structures, but on the strategies of the partners, the collective games to which these strategies correspond, and the rules of regulation which govern these games. Interorganizational systems and single organizations are both more or less stable and structured networks of social and human relationships.

### **Power and Bargaining**

The basic postulate on which systems analysis should be founded is that all systems structure and regulate themselves around the power relations established between their members and their units.

Systems are based on interdependence. This interdependence is not symbolic, it is built on relations of concrete dependence which cannot be understood as the inevitable consequence of objective situations, but rather as the result of constantly questioned relations of power. To analyze these relationships, one should not measure the resources as an accountant, but make a qualitative assessment of the actions open to the partners and of the dynamics of their games. It is equally indispensable to focus on the relationship as such and not on each partner's respective power. Even if one partner appears completely to dominate the other, the dependence remains reciprocal—no matter how absolute the right of life and death is held by masters over their slaves. Masters are dependent on their slaves' survival in order to retain lordship over them.

Literature in the field has rightly underscored the fact that the relation of the organization to its environment is a relationship of dependence (Levine and White, 1961; Yuchtman and Seashore, 1967; Thompson and McEwen, 1958). Yet studies usually describe and classify the resources of power employed in this relation more than show the mechanisms of the game between the partners. They cling to the objective resources and have been content to accept the channel and the nature of the exchanges as a valid empirical indicator (Aldrich, 1974).

Such an approach is lacking. The nature of the relationship does not depend on the resources of each player, but on the powers which they use in the game. The approach suggested here demonstrates the significance of the relevant capacities of the players by exploring the areas of uncertainty affecting the respective aims and the common objectives which they can control. Such general, but empirical, criteria of analysis on the one hand can help measure the forces or, more precisely, the relevant means of actions. On the other, it can make it possible to understand and describe the maneuvers of the players and the dynamics of their relationships (Crozier, 1964, 1974).

Power exercised in a relationship cannot be deduced by an objective analysis of opposing forces. But one can appreciate its weight and its meaning from the feelings and attitudes of the protagonists.

These feelings can be found just as readily with in-depth interviews as through questionnaires of attitudes. They reflect in detail the real power situation, the sensitive points of the relationship, and the characteristics of the game. Those interviewed do not abstain from rationalizing their feelings, yet their rationalizations are particularly revealing since they can be compared with those of colleagues and partners in the same and opposite situations.

Thus, by using their statements one can deal with rough, but relatively faithful, indicators of the direction of the power relations between the different partners and of the problems around which these relations crystallize. Yet, to analyze power or the power relationship is not an end in itself; by considering this relationship, one has to describe the players' strategy and understand the games in which they develop these strategies.

The information gathered on the conflicts of power and the problems around which they crystallize make it possible to understand the manner in which the partners bargain and how transactions take place between them. From this one can infer their conscious or unconscious strategy. Then, by using this knowledge to analyze major decisions and by comparing the cross-relations data with the objective data of the situation, one can reconstruct the explicit or implicit game which underlies the strategies and relationships.

These games can differ greatly and not correspond at all to formal appearances. Players deriving strength from the authority legally vested in them can find themselves blocked in games in which their best strategy consists of minimizing their losses. In general, of course, powerful partners have

more chances of winning because they deal with various resources and can thus engage in a larger number of different games. Being better able to allot their stakes and run more risks with reason, they have more chance of making a much better than average score in a long set of contests, and thus win in the end. The opportunities open to each partner for bargaining and cooperation tend to stabilize the games into relatively permanent structures. The partners, however, remain vulnerable to the constant changes that affect the uncertainties they control. The possibilities of reopening the question are proportionately much greater when these controls are less rigorous.

One seeks to understand the strategy of each partner by analyzing the juxtaposition of their converging and contradictory attitudes and feelings. One postulates that these strategies are rational in relation to the situation. From the objective data about the situation and the way each of the players experiences it, one tries to reconstruct what their game is practically and what the rules are that can fit the strategies which have been hypothesized (Crozier, 1972).

### **Games, Regulations, and System**

If a system is postulated as existing, there has to be a type of regulation which can be either explicit or implicit. In other words, relationships between parts of the system are governed by collective rules which are relatively stable. No part can develop its own game independently from the way the other parts play their game and from the general rules. Individual behavior is constrained by a set of rules. At the system level, it is therefore possible to find the same rules that govern the power relations as at the interpersonal level. Some behaviors are penalized. Some strategies cannot be played. The gains and losses are, if not determined, at least defined and limited by the existence of social rules. Therefore, each member of the system is in a position to work out a rational calculus and to formulate a strategy.

In sociological terms, a system is characterized by the means of regulation it uses. It is a system inasmuch as it influences members by regulating the games they play. From that point of view, regulation is its mode of government, with "government" used not in its legal or formal meaning, but in its cybernetic sense.

This study uses system analysis to understand the games which are played at the bottom of the system. Then it proceeds to discover and reconstruct the more general mode of regulation through which these games are articulated to one another. Finally, it tries to ascertain the basic mode of government and the values that make it possible for the system to exist by legitimizing these rules.

This approach, a progressive discovery of the latent structure of a system, will raise a number of basic problems. There is first a problem of rationality. This problem is logically the same as encountered in dealing with the strategies at the operational level. Empirically, however, it is much more difficult because the analyst must enter into a much less constrained field in which the freedom of individuals in relation to the roles they assume is much greater. The rationality of

individuals does not necessarily correspond to the rationality of the group they represent. Not only do individuals have many more possible strategies within the game in which they are engaged, but they also have several games available to them. They can consider committing themselves more or less to one or the other. The reasoning used in discovering the meaning of the game—by postulating that the meaning of the game is what makes the various strategies equally rational—proves to be much more difficult to put into action. By observing the recurrences and by analyzing the possible impact of change it is possible to reduce the relative confusion imposed by the fluidity of the game and the freedom of the individual.

A second problem is raised by the lack of a homologous relationship between the games one finds at operational levels and the game which regulates the system. There is a great temptation to extrapolate from the repetition of the same games at the operational level a general model of regulation for the whole. This conclusion is not totally false and the empirical analysis of the system as described here shows that there are indeed certain similarities. But this does not mean that the game which regulates the whole is of the same sort as the games analyzed at the bottom. For example, it was shown that the regulation of a system in which games at the operational level are of a defensive bureaucratic nature operates according to a much more open model. People in that model have to assume risks directly. Conversely, the basic cross-regulation system—analyzed here at the local level—may be regulated by a charismatic or authoritarian structure at the top.

### The Example of French Local Government

Using the case of the French local government, the meaning of the present results, the relevance of the mode of reasoning suggested and the uncertainties it still involves can be understood. The French system is characterized by the small number of participants, the permanence of their roles, the secrecy of their relationships, and the closed nature of the whole with respect to outsiders and the public in general. These characteristics can be deduced neither from the formal legal structure of administrative and political powers, nor from the values professed within the political community. They can also be derived from the motivations of any of the parties involved in the political game. Nevertheless, these characteristics appear to be perfectly rational if one admits that the games played by the actors really exist. It is rational and even very democratic for mayors of small towns to monopolize the relationship with administrative partners and not inform their councils, since in this game, the major trump they hold is the unity of the community in relation to the diversity of its partners. It is rational to select someone who occupies a plurality of offices. Untimely publicity or intrusion of new partners, far from reinforcing the mayors, can only put in question the privileges they have been able to accumulate and which alone allow them to succeed in the game.

Obviously, individual strategies can be found to be different, but their rationality can be discovered only after the rationality of the collective game has been well analyzed. It is in terms of this regularity that the exception becomes meaningful and

can be hypothesized as being rational in view of the specificity of the situation.

Mayors who play on the openness and participation of citizen groups can maintain their influence only inasmuch as they find partners in the games they play at other levels who will reward their initiatives. This may be possible if they represent for them a useful exception. Their behavior then appears to be rational, even if they act in opposition to traditional strategies. But for such a behavior to predominate, the general regulation of the system, that is the whole system, must change.

Several rational strategies are always possible in most of these games, especially inasmuch as several games interpenetrate or the personalities of the partners enter into the picture. Even if the structure of the game presupposes that the relation between mayors who hold only one office and mayors who hold a plurality of offices is relatively loose, it can become very rational for the first ones to give their allegiances to the second. They may do so to reinforce their own power when dealing with the local state bureaucrat or to seek in their turn to hold another office, always on the condition that they find themselves in a sufficiently powerful bargaining position. Besides, the conditions for this bargaining and its rationality will depend directly on the structure of the collective game they will be contradicting.

At the level of regulation of the whole, there are two main characteristics which at first glance are not at all congruent. On the one hand, there is the generality of cross-controls, a phenomenon according to which the regulation of a relationship or an organizational channel is performed by the members of another organization which is itself eventually regulated by the former. On the other hand, there is the decisive importance of exceptions to the prevalent game. This means that the really powerful are always those who can escape the rules of the system and can therefore exert pressure more directly on its nodal points where the decisions have to be taken.

Thus, the way the whole system is regulated cannot easily be extrapolated from an analysis of the basic model of cross-controls, since those who win in the game of regulating the whole are those who manage to elude the dominant mode of regulation. As a result, one can formulate the following hypothesis: the essential property of the system is that its regulation relies on the existence of a double series of games which evolve naturally from the uncertainties of interdependent activities (Crozier, 1964). But this reasoning the opportunities offered by the difficulties of articulation between the intersecting games at the operational level. This is achieved only by manipulating exceptions to the predominant rules. But these exceptions are at the same time recognized and coded and they themselves tend to develop into other games. They move toward another kind of rigidity eventually requiring some new exception.

These hypotheses cannot be generalized. But they draw attention to the crucial meaning of new phenomena requiring further exploration both within interorganizational systems and within organizations which, under the circumstances, can

be considered as special cases in the class of general systems.

First to consider is the phenomenon of control or cross-regulation. The prevalent thinking about the problem of authority or government within organizations and systems has always tended to focus on hierarchical relations. Is it not sensible to put this logic into a question? Haven't researchers been misled by the prevalent hierarchical ideology, even when they were fighting it, and haven't they accepted a priori a rough equivalence between hierarchic authority and social control? Maybe these phenomena of cross-relations which constrain the players narrowly without putting them in a position of direct dependence are finally much more important than hierarchical authority.

A second phenomenon, regulation by exception, should attract further attention. By successfully escaping the rules that govern its fundamental games, one is able to participate in the government of a system. Such a hypothesis calls for a reconsideration of the relations between the formal and the informal. Traditional reasoning on this subject has been that the field is structured by the formal and that the informal that develops out of the loopholes constitutes a more or less tolerable exception in the formal structure. Later, human relations reasoning developed around the view according to which the real life of an organization lies in the informal and that the formal is but a superstructure of little importance. Another kind of reasoning should be applied to the analysis of organizations. It consists of considering the formal as a necessary response to the pressures of informal blackmail, a means of governing by structuring and stabilizing the power games which evolve naturally from the uncertainties of interdependent activities (Crozier, 1964). But this reasoning should be completed by a complementary distinction. Instead of opposing only the formal and the informal, one should also oppose the regular game which is made up of a formal structure and an informal game, and a second game for initiates which grows out of the first. That is the game of the exception through which one can solve the extremely difficult problem of the relationships between the games. It is through the weaknesses in their models of social control that most systems govern themselves and solve their contradictions. Above all, systems analysis should endeavor to discover the properties and consequences of this second kind of game.

Such hypotheses contradict the approach of traditional institutional analysis, according to which actual behaviors can be deduced from the history of the structures, the history of the mentality, or from the recurrent renovation of formal institutions by opposition and revolt.

### The Problem of Change

The problem of change is generally not taken into account in the literature on the organization-environment relations or even interorganizational systems. In appearance, the line of thought inspired by the structural contingency model seemed to be able to offer an opening for understanding change. If sufficient correlation is found between a certain kind of environment and certain kind of organizational structure, one can deduce a natural tendency toward evolution and sources

of change. This mode of reasoning is tempting, but dangerous. The regularities between certain states of the environment and certain states of the organization and its performance do not make it possible to propose general laws. At most, they suggest normative prescriptions. This orientation was emphasized by such authors as Burns and Stalker (1961), and Lawrence and Lorsch (1967). But behind the normative aspect, there remains always a certain kind of determinism. An environment of a certain nature calls for an organization of a certain nature which alone will give the best possible results. If at any given moment an organization does not have the structural properties corresponding to the demands of the environment, it is assumed to be in a very bad position and to have to change its nature unless it accepts the risk of being wiped out. But generalized in this way, the logical propositions of the structural contingency model do not resist empirical verification. As suggested by Pennings (1974), organizations succeed very well when transgressing the laws of the model and the normative conclusion cannot be supported by any empirical determinism. These studies have certainly brought about a decisive change in the dominant universalistic model. But their authors and even those such as Perrow (1972) and Thompson (1967) free themselves with difficulty from its logic. What they fail to do is consider the organizational or systemic capacity to solve the problems of an organization's relations to its environment as a complex human phenomenon and not as a dependent variable or a subject matter for decision making.

A model for government—based on a double series of contradictory games with a dominant mode of defensive strategies but with a possible opening for the offensive strategies of a small group of leaders who have been selected because of their intuitive comprehension of the system—cannot be considered as determined by an environment which it is structuring. Nor is it the result of deliberate decisions developed with the help of wise consultants. It represents a collective capacity built up through the trials and errors of history to solve the problem of the internal and external environments.

One might argue that the example of local public institutions in France is a very specific one and that organizations with clearer aims and which are more directly subject to the sanction of the environment do not develop such complexities. This issue cannot be solved too hastily. For complex as well as for more simple organizations, the decisive variable for their activity is their collective capacity. In other words, the set of games played in an organization or in an interorganizational network, and their regulation, constitute a system. It is this system which has the capacity to find solutions to the problems of the environment. A system dominated by defensive or bureaucratic games has greater difficulty in solving problems raised by the adaptation to a turbulent environment. But if the regulation of these games is put into effect in such a manner, charismatic figures will emerge and offer intuitive solutions to the problem of change and adaptation. Therefore, any normative advice should take into account the learning capacity of the system. Under certain conditions, for instance, if the participants are not able to play the new games im-



posed by these transformations, it is preferable to reinforce an integrating centralization mechanism around individuals and groups capable of rapid adaptation, rather than to transform the whole structure so that it fits the nature of the problem.

A new orientation is indispensable in a review of a theory of organizations. Very little is known about the process of collective learning of new models of human relations. What is known of learning at the group level—and which is itself quite speculative with respect to what is known about individual learning—is often extrapolated too hastily to the level of the organization as such. The literature on change has, until now, hardly taken into account the systemic dimension of organizations (Bennis, 1966; Bennis, Benne, and Chin, 1964). The time has come to progress in such a direction. Unfortunately, experimentation at that level is very difficult. An alternative way to proceed involves the study of reforms continuously attempted, and rarely successful, in the field of public government. Over and beyond the difficulties it raises, such a strategy requires a much less formalized approach and puts into question, at least partially, progress made in the field of quantification. It has the advantage of remaining close to the experimental method and it may produce cumulative conclusions on the conditions, possibilities, and limitations of collective learning processes in organizations and in organized systems.

**Michel Crozier** is Director of the Center for Sociology of Organizations at the National Center of Scientific Research in Paris, France. **Jean-Claude Thoenig** is a professor of sociology at the Federal Institute of Technology in Lausanne, Switzerland.

## REFERENCES

- Aiken, Michael, and Jerald Hage**  
1968 "Organizational interdependence and intra-organizational structure." *American Sociological Review*, 33: 912–930.
- Aldrich, Howard**  
1974 *The Environment as a Network of Organizations: Theoretical and Methodological Implications*. Toronto: VIII World Congress of Sociology.
- Bennis, Warren**  
1966 *Changing Organizations*. New York: McGraw-Hill.
- Bennis, W., K. D. Benne, and R. Chin**  
1964 *The Planning of Change*. New York: Holt Rinehart and Winston.
- Benson, Kenneth**  
1974 *The Organizational Network as a Political Economy*. Toronto: VIII World Congress of Sociology.
- Burns, T., and G. M. Stalker**  
1961 *The Management of Innovation*. London: Tavistock.
- Clark, Burton**  
1965 "Interorganizational pattern in education." *Administrative Science Quarterly*, 10: 224–237.
- Crozier, Michel**  
1964 *The Bureaucratic Phenomenon*. Chicago: Chicago University Press.  
1972 "The relationship between micro and macro sociology." *Human Relations*, 25: 239–251.  
1974 *The Stalled Society*. New York: Viking Press.
- Crozier, Michel, Erhard Friedberg, Catherine Grémion, Pierre Grémion, Jean-Claude Thoenig, and Jean-Pierre Worms**  
1975 *Où va l'Administration Française*. Paris: Editions d'Organisation.
- Dill, William**  
1958 "Environment as an influence on managerial autonomy." *Administrative Science Quarterly*, 3: 409–443.
- Duncan, Robert**  
1972 "Characteristics of organizational environments and perceived environmental uncertainty." *Administrative Science Quarterly*, 17: 313–327.
- Emery, F. E., and E. L. Trist**  
1965 "The casual texture of organizational environment." *Human Relations*, 13: 21–32.
- Evan, William**  
1966 "The organization-set." In James D. Thompson (ed.), *Approaches to Organizational Design*. Pittsburgh: University of Pittsburgh Press.

**Grémion, Pierre**

- 1966 "Introduction à une étude des systèmes politico-administratif local." *Sociologie du Travail*, 3: 51–73.
- 1976 *Le pouvoir pèrithérique*. Paris: Le Seuil.

**Grémion, Pierre, and Jean-Pierre Worms**

- 1970 *Les Institutions Régionales et la Société Locale*. Paris: Center for Sociology of Organizations.

**Hall, Richard**

- 1972 *Organizations, Structures and Processes*. Englewood Cliffs, N.J.: Prentice-Hall.

**Katz, Daniel, and Robert Kahn**

- 1966 *The Social Psychology of Organization*. New York: Wiley.

**Kesselmann, Mark**

- 1966 *French Local Government: The Politics of Consensus*. New York: Knopf.

**Lawrence, Paul, and Jay Lorsch**

- 1967 *Organization and Environment*. Boston: Harvard University Press.

**Levine, Sol, and Paul White**

- 1961 "Exchange and interorganizational relationships." *Administrative Science Quarterly*, 10: 583–601.

**Litwak, Eugene, and Lydia Hylton**

- 1962 "Interorganization analysis: a hypothesis on coordinating agencies." *Administrative Science Quarterly*, 10: 395–420.

**Pennings, Johannes**

- 1974 *Environment, Structure and Performance of Complex Organizations*. Toronto: VII World Congress of Sociology.

**Perrow, Charles**

- 1972 *Complex Organizations: A Critical Essay*. Glenview, Ill.: Scott Foresman.

**Thoenig, Jean-Claude**

- 1973 *L'Ere des Technocrates*. Paris: Editions d'Organisation.
- 1975 "La relation entre le centre et la peripherie en France." *Bulletin de l'Institut International d'Administration Publique*, 36: 77–124.

**Thompson, James D.**

- 1967 *Organizations in Action*. New York: McGraw-Hill.

**Thompson, James D., and William MacEwen**

- 1958 "Organizational goals and environment." *American Sociological Review*, 23: 23–31.

**Turk, Herman**

- 1970 "Interorganizational networks and urban society." *American Sociological Review*, 35: 1–19.
- 1973 "Comparative urban structure from an interorganization perspective." *Administrative Science Quarterly*, 18: 37–55.

**Warren, Roland**

- 1967 "The interorganizational field as a focus of investigation." *Administrative Science Quarterly*, 12: 396–419.

**Weick, Karl**

- 1969 *The Social Psychology of Organizing*. Stanford, Ca.: Addison Wesley.

**Yuchtman, Ephraim, and Stanley Seashore**

- 1967 "A system resource approach to organizational effectiveness." *American Sociological Review*, 32: 891–903.

**Worms, Jean-Pierre**

- 1966 "Le préfet et ses notables." *Sociologie du Travail*, 3: 249–275.

Copyright of Administrative Science Quarterly is the property of Administrative Science Quarterly and its content may not be copied or emailed to multiple sites or posted to a listserv without the copyright holder's express written permission. However, users may print, download, or email articles for individual use.